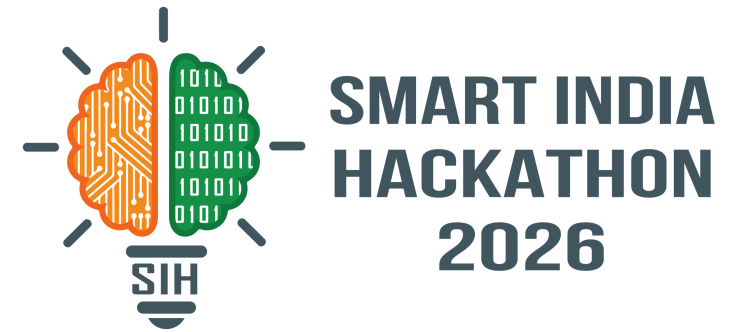


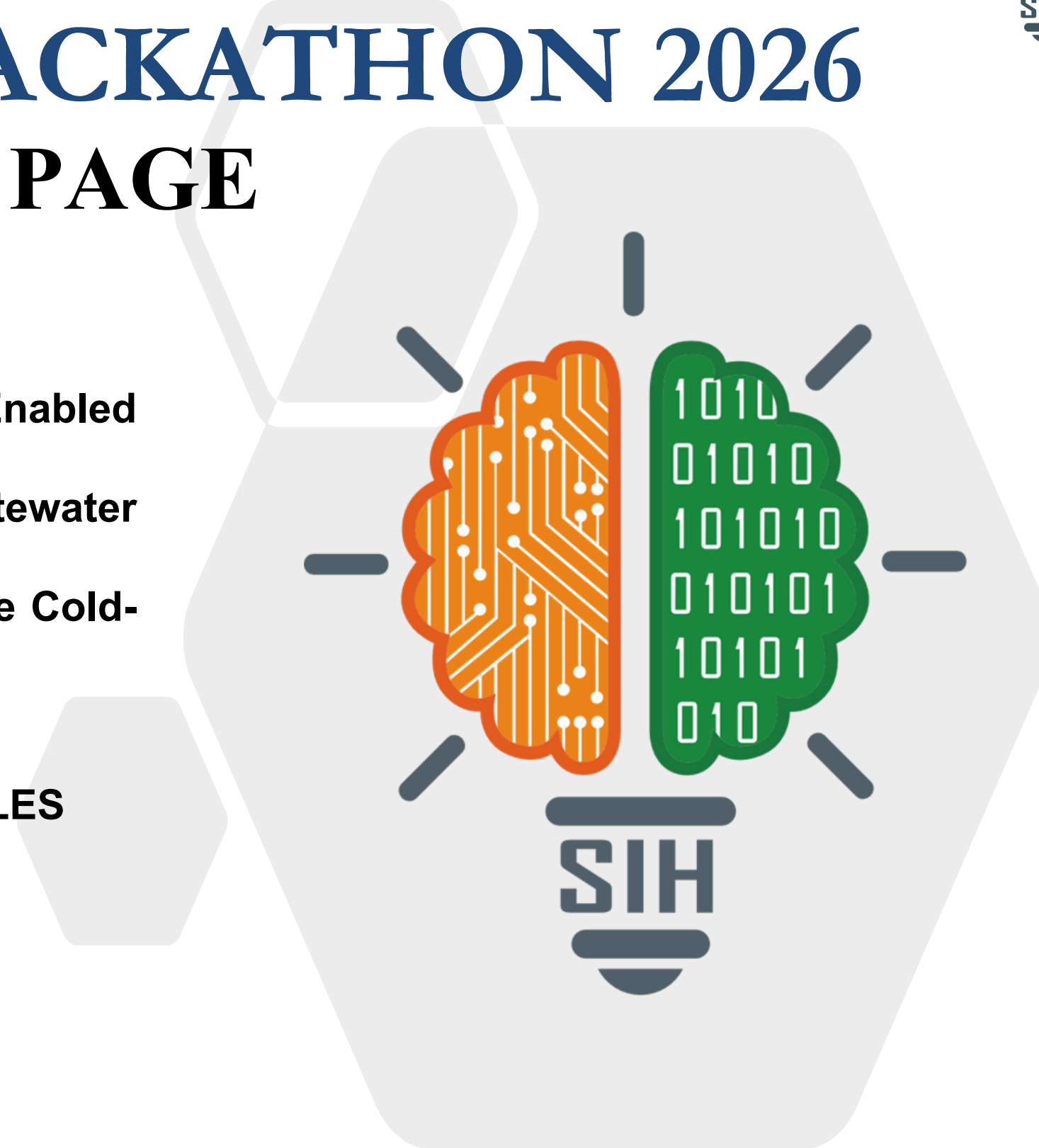
VIT Bhopal University

INTERNAL HACKATHON 2026

TITLE PAGE



- Problem Statement Title-An Edge-AI & Blockchain-Enabled System for Pre-Outbreak Detection — Coupling Wastewater Pathogen Early Warning with Tamper-Proof Bio-Sample Cold-Chain Tracking
- Theme- DIGITAL BIO-SAMPLES AND SANITARY SAMPLES
- PS Category- Software+Hardware
- Team ID-VITBSIH26-474
- Team Name : Team Achilles



AarogyaKavach:

- AarogyaKavach is a dual-pillar IoT ecosystem for public health
- Pillar 1:** Asha Smart Box + **Pillar 2:** Sewer Node = **Central AI Cloud Brain**

Problems We Address

1) Biological Sample Loss
zero ice box monitoring

2) Detection Delay
hospital only detection

3) Slow Response of health teams

Our solution

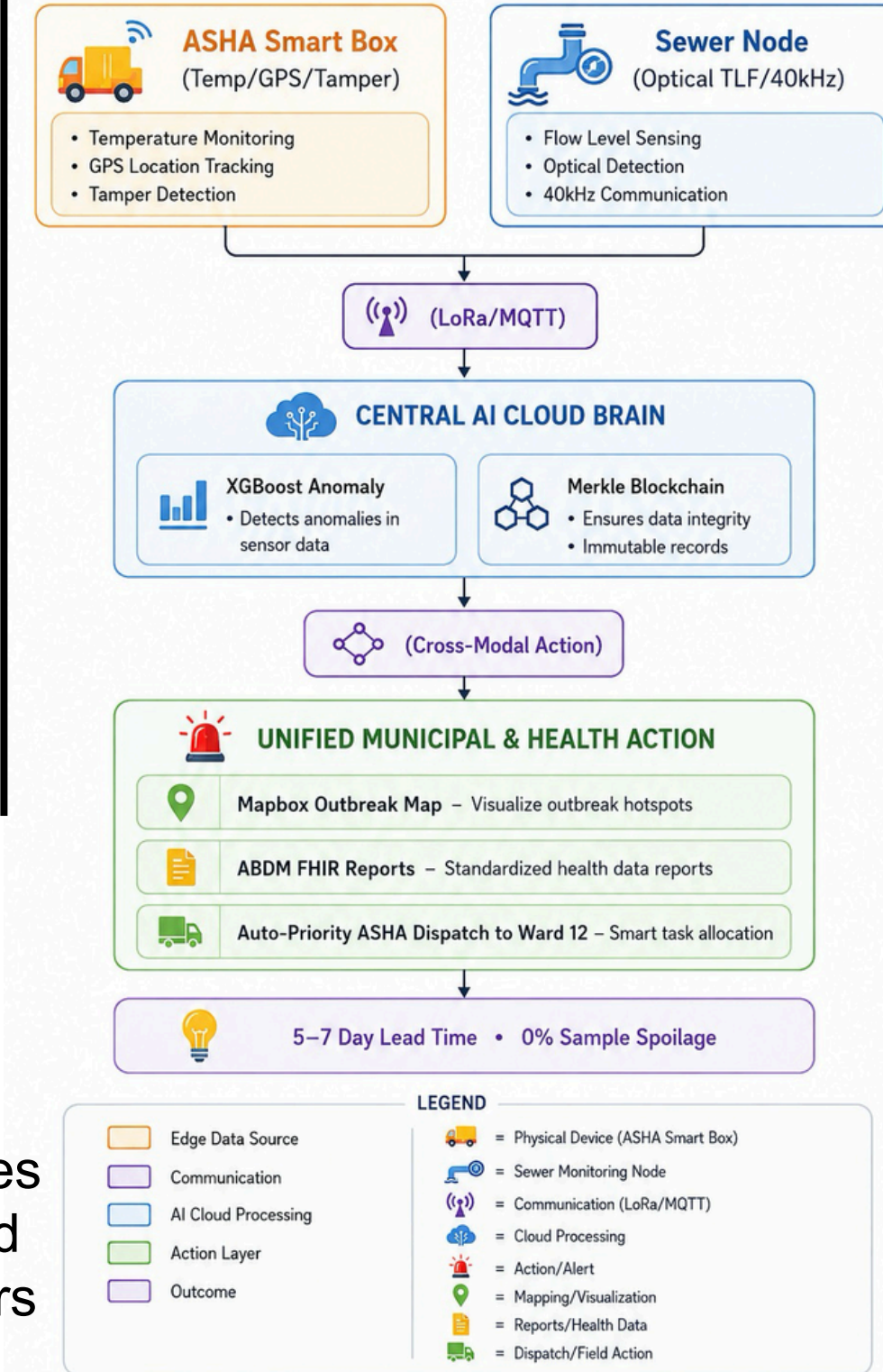
1) Smart box-tracks temperature, GPS, and tampering; sends WhatsApp/buzzer alerts if ice melts ($>8^{\circ}\text{C}$) or lid is breached

2) Sewer Node uses optical sensing + 40 kHz self-cleaning (zero chemicals) to spot viral surges in wastewater

3) AI Cloud Brain runs XGBoost anomaly detection + Merkle blockchain for secure data

4) Unified Action Dashboard shows outbreak maps, generates ABDM FHIR reports and dispatches ASHA workers to priority wards

COMPLETE SYSTEM FLOWCHART



TECHNICAL APPROACH

TECH-STACK:

HARDWARE:

- ESP32-S3, DS18B20, DHT22, Neo-6M GPS, MPU-6050, Hall Sensor, SX1276 LoRa, UV Optical Sensor, pH Sensor

SECURITY:

- SHA-256 Hashing, Merkle Tree Verification, Ed25519 Digital Signatures, ESP32 eFuse Key Storage

INTEGRATIONS:

- ABDM FHIR R4, IHIP Alert Format, Smart Cities ICCC (GeoJSON)

BACKEND:

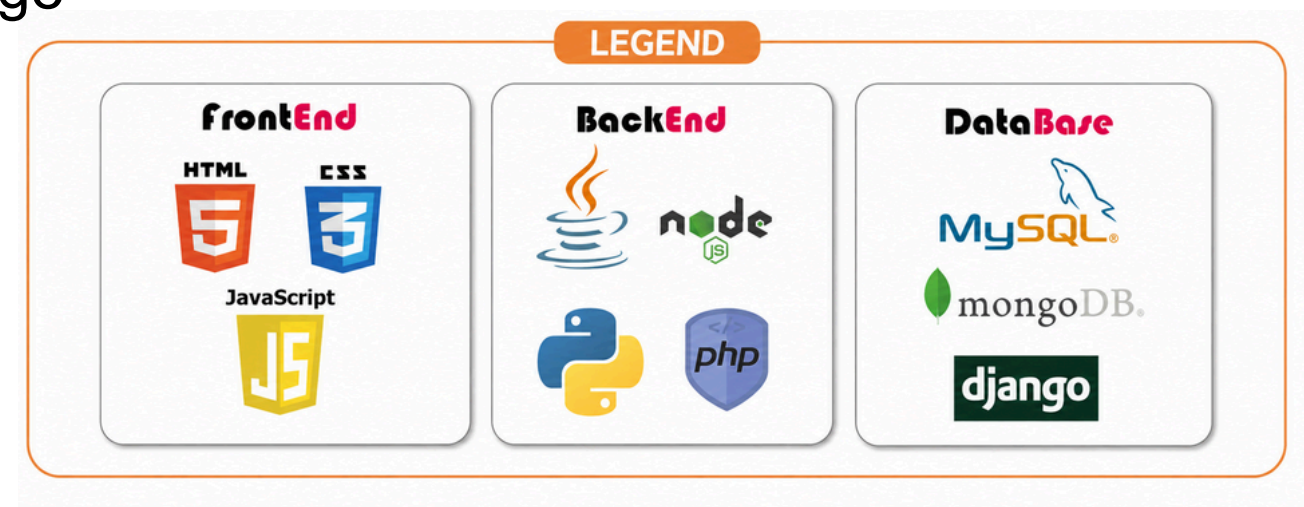
- Python, FastAPI, MQTT, PostgreSQL, TimescaleDB, Redis Streams

WEB & MOBILE APPLICATION:

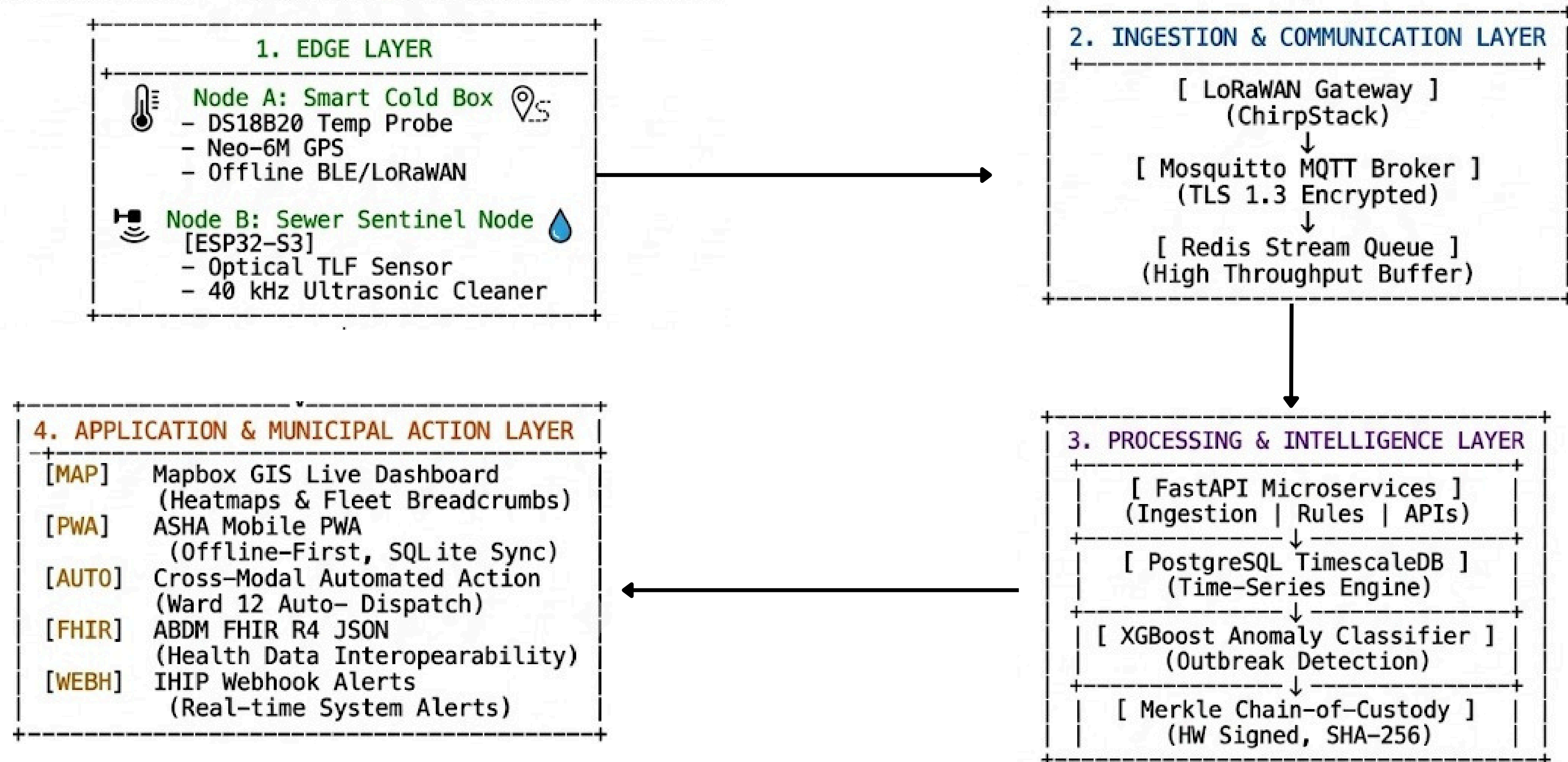
- React 18, React Native (PWA), Mapbox GL, SQLite (Offline Sync)

AI & ML:

- Gradient Boosted Trees, Ward Priority Scoring, 7-Day Outbreak Forecasting



AAROGYAKAVACH – ASCII ARCHITECTURAL WORKFLOW



1

FEASIBILITY ANALYSIS

Technically sound · Cost-viable · Field-deployable

- **BOM Cost — < ₹1,850 / unit**
- **Standby Battery — 18 Days**
- **Anti-Fouling Tech — 40 kHz Ultrasonic Cavitation**
- **Connectivity — LoRaWAN 865 MHz**
- **Field Sync — Offline Sync < 2s**

Verdict: ₹1,850 BOM + 18-day autonomy makes 1,000+ village-level nodes viable under Jal Jeevan Mission / smart-city budgets.

2-3

POTENTIAL CHALLENGES & ENGINEERING STRATEGIES

CHALLENGE	ENGINEERING STRATEGY
Sewer Sensor Biofouling: Organic/microbial buildup blinds pathogen sensors within days.	Automated 40 kHz Ultrasonic Cavitation self-clean cycle every 6 hrs → zero manual maintenance, restores “>95%” accuracy.
False-Positive Pathogen Alarms: Single-sensor drift/noise triggers unnecessary outbreak alerts.	Edge-AI triple-sensor consensus voting + Bayesian confidence filter — cuts false positives to < 2% before blockchain logging.
Deep Rural 4G Dead Zones: No cellular coverage at remote sample-collection & sewer nodes.	LoRaWAN 865 MHz mesh backhaul with Offline Sync < 2s buffering — auto-uploads the instant a node re-enters range.



IMPACTS

- **Tamper-Proof Diagnostic Integrity:** Blockchain-chained logs (SHA-256 + Ed25519) eliminate false negatives from broken cold chains or compromised samples.
- **Community-Level Early Warning.**
- **Smart Field Dispatch ML priority scoring auto-routes** ASHA workers to highest-risk wards instead of fixed schedules.
- **Government-Ready Interoperability.**
- **Scalable Municipal Costing.**

BENEFITS

- **5–7 Days Earlier Detection.**
- **~70% Lower Cost vs. Pharma Trackers.**
- **100% Cold-Chain Accountability.**
- **Zero Dead-Zone Dropouts.**
- **Precision Resource Deployment.**

Projected outcomes



5–7 Days

Earlier outbreak warning



2–8°C

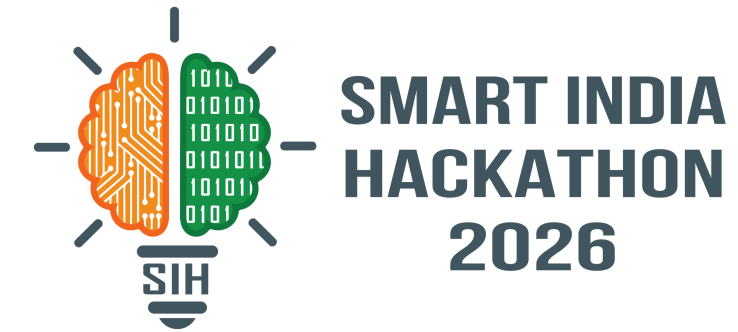
Continuous cold-chain monitoring



48–72 hrs

Offline operation capability

RESEARCH AND REFERENCES



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- 1. WHO Environmental Surveillance for SARS-CoV-2 (2022 Interim Guidance)
https://www.who.int/publications/i/item/WHO-2019-nCoV-Surveillance_Wastewater-2022.1
- 2. WHO Environmental Surveillance Guidance (Updated 2023)
<https://www.who.int/publications/i/item/9789240074653>

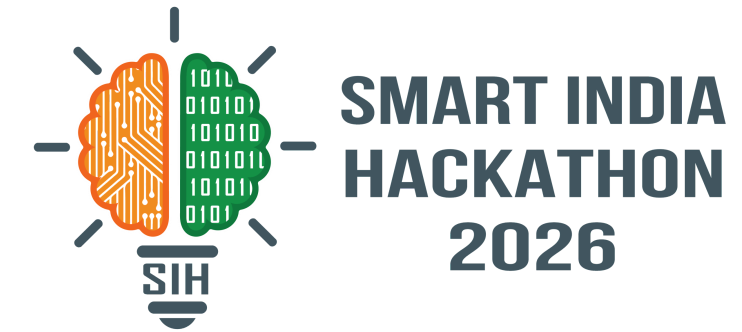
2. Cold-Chain Integrity & Vaccine Logistics

- 3. WHO PQS (Performance, Quality & Safety) Standards
https://apps.who.int/immunization_standards/vaccine_quality/pqs_catalogue/
- 4. MoHFW – National Cold Chain Management Information System (NCCMIS)
<https://nccmis.gov.in>

3. National Digital Health Standards

- 5. NRCeS – HL7 FHIR R4 Implementation Guide for ABDM
<https://nrce.in/ndhm/fhir/r4>
- 6. NRCeS Digital Health Resources
<https://nrce.in>

TEAM DETAILS



MENTOR DETAILS:GUNJAN ANSARI(SCOPE)

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TEAM LEADER NAME :BIPANCHI KALITA

TEAM MEMBER 1:TUSHAR BURLA CHANDRAMOULI

TEAM MEMBER 2:ARPIT GUPTA

TEAM MEMBER 3: SARVAGYA SHRIVASTAVA

TEAM MEMBER 4:ARUN TOMAR

TEAM MEMBER 5: SANIDHYA PORWAL

